1 Here is a list of 10 numbers.

2 3 4 4 4 5 6 6 7 7

(a) Work out the range.

$7 - 2 = 5$

5 .....
(1)

(b) Find the mode.

4 .....
(1)

(c) Calculate the mean.

$2 + 3 + 4 + 4 + 4 + 5 + 6 + 6 + 7 + 7 = 48$

$\frac{48}{10} = 4.8$

4.8 .....
(2)

(Total for question 14 is 4 marks)

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2 Here is a list of 5 numbers.

4 6 9 10 11

(a) Work out the range.

$11 - 4 = 7$

7 .....
(1)

(b) Write down the median.

9 .....
(1)

(c) Calculate the mean.

$4 + 6 + 9 + 10 + 11 = 40$

$\frac{40}{5} = 8$

8 .....
(2)

(Total for question 2 is 4 marks)

3 Here are the heights, in cm, of 8 people.

155 171 164 171 167 188 190 151

(a) Work out the range.

190 − 151

39 cm
(1)

(b) Find the mode.

171 cm
(1)

(c) Calculate the mean.

155 + 171 + 164 + 171 + 167 + 188 + 190 + 151

= 1357

$\frac{1357}{8}$ = 169.625

169.625 cm
(2)

(Total for question 3 is 4 marks)

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4 Here are the weights, in grams, of 6 potatoes

150 129 125 133 144 105

(a) Work out the range.

150 − 105 = 45

45
(1)

(b) Work out the median weight.

105 125 129 133 144 150

↑

$\frac{129 + 133}{2}$ = 131

131 g
(2)

(Total for question 4 is 3 marks)

5 Here are six cards. Each card has a number on it.

19	7	11	8	15	15

(a) Work out the range of the numbers on the cards.

$19 - 7 = 12$

12

(1)

(b) Work out the mean of the numbers on the cards.

~~19 + 7 +~~

$19 + 11 + 15 + 15 + 8 + 7 = 75$

$\frac{75}{6} = 12.5$

12.5

(2)

(Total for question 5 is 3 marks)

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6 Here is a list of 10 numbers.

1 4 4 5 6 8 11 11 11 14

(a) Work out the range.

$14 - 1 = 13$

13

(1)

(b) Find the mode.

11

(1)

(Total for question 6 is 2 marks)

7 Here are seven cards. Each card has a number on it.

12	5	10	18	12	11	9

(a) Work out the range of the numbers on the cards.

$18 - 5 = 13$

13
(1)

(b) Work out the median of the numbers on the cards.

5 9 10 11 12 12 18

11
(2)

(c) Work out the mean of the numbers on the cards.

$5 + 9 + 10 + 11 + 12 + 12 + 18 = 77$

$\frac{77}{7} = 11$

11
(2)

(Total for question 18 is 5 marks)

8 Here is a list of numbers.

8 6 4 5 9 8

(a) Work out the median

4 5 6 8 8 9

$\frac{6+8}{2} = 7$

7

(2)

Here are six cards.
There is a number on each card.
Two of the numbers are hidden.

4	5	?	6	3	?

The mode of the six numbers is 4
The mean of the six numbers is 5

Work out the two numbers that are hidden.

There must be at least one more 4.

4 5 4 6 3 x

The numbers must add to $5 \times 6 = 30$

$4 + 5 + 6 + 3 + 4 = 22$

$30 - 22 = 8$

4 , 8

(2)

(Total for question 8 is 3 marks)

9 Here is a list of numbers.

14 19 15 20 11 14 19

(a) Find the range

20 − 11 = 9

9

(2)

(b) Calculate the mean

14 + 19 + 15 + 20 + 11 + 14 + 19 = 112

$\frac{112}{7} = 16$

16

(2)

Andrew says,

"The median is the middle number, so the median is 20."

11 14 14 15 19 19 20

(c) Andrew is incorrect, explain why.

The numbers have to be ordered first.

The median is 15.

(1)

(Total for question 9 is 5 marks)

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10 Here is a list of seven numbers.
One of the numbers is hidden.

11	6	7	10	7	9	?

The mean of the numbers is 9.

Find the value of the hidden number.

7 × 9 = 63 The numbers must add to 63

11 + 6 + 7 + 10 + 7 + 9 = 50

63 − 50 = 13

13

(Total for question 10 is 2 marks)

11 The mean of eight numbers is 41.
The mean of two of the numbers is 29.
Work out the mean of the other six numbers.

$8 \times 41 = 328$ All numbers total 328

$2 \times 29 = 58$ Two numbers total 58

$328 - 58 = 270$

$\frac{270}{6} = 45$

45

(Total for question 11 is 3 marks)

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12 Mark ran a~~t~~ mean distance of 13.2 km in five days.

The next day Mark ran 20 km.

Find the mean distance Mark ran in the six days.

$13.2 \times 5 = 66$ km in 5 days

$66 + 20 = 86$ in 6 ~~s~~ days

$\frac{86}{6} = 14.\dot{3}$ km

$14.\dot{3}$ km

(Total for question 12 is 3 marks)